

Windows Authentication Investigation Report

Objective: Reviewing Windows Security event logs to identify authentication patterns, investigate failed logon attempts, and distinguish normal activity from events requiring escalation.

REPORT DETAILS

Analyst: Lydia Azra

Date: [4 September 2026]

Host Analysed: [AZRA-LAPTOP / Windows 11]

Investigation Period: [28 August 2026 – 4 September 2026]

Tools Used: Windows Event Viewer, Event Log (Security)

EVENT ID REFERENCE

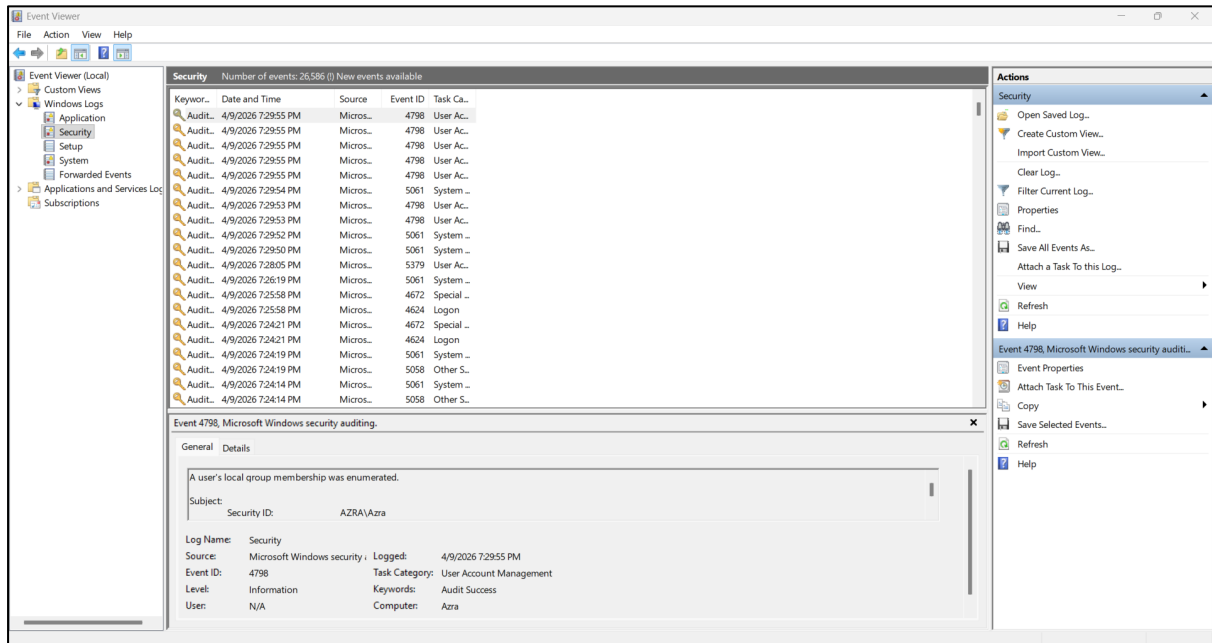
The following Windows Security Event IDs were used as the basis for this investigation:

Event ID	Event Name	What It Means / What to Look For
4624	Successful Logon	Note the Logon Type (2=interactive, 3=network, 10=remote), username, source IP, and timestamp.
4625	Failed Logon	Record username, failure reason (wrong password vs unknown user), source IP. Repeated failures = brute force indicator.
4740	Account Lockout	Which account was locked, from which machine. Often follows repeated 4625 events. Investigate source.
4648	Logon with Explicit Credentials	User authenticated using different credentials (e.g. runas). Can indicate lateral movement or privilege escalation attempts.

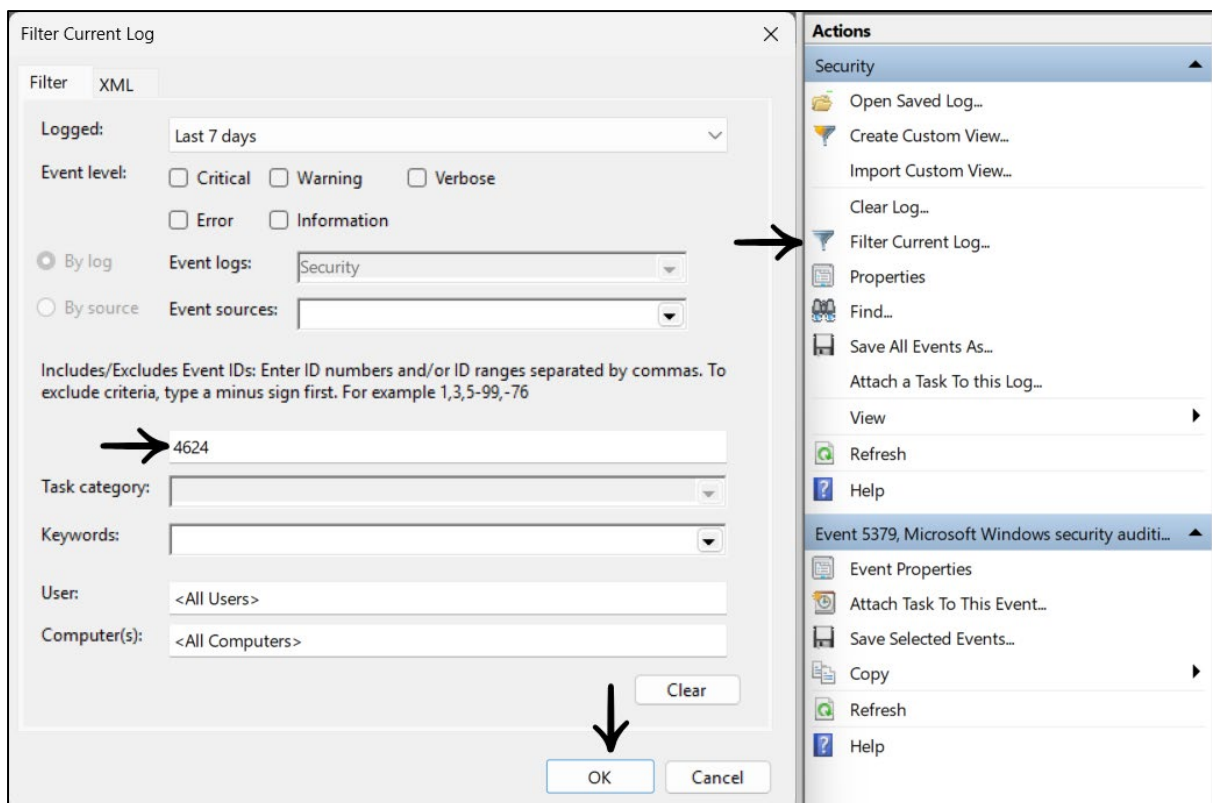
HOW TO ACCESS SECURITY LOGS

Using Event Viewer (GUI)

- Press Win + R → type eventvwr.msc → press Enter
- Navigate to: Windows Logs → Security
- Use Filter Current Log (right panel) → enter Event IDs: 4624, 4625, 4740, 4648



Event Viewer – Windows Logs > Security (unfiltered)



Filter Current Log > enter event ID

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 26,538 (0) New events available

Filtered: Log: Security; Source: ; Event ID: 4624 Date Range: Last 7 days. Number of events: 991

Keyword...	Date and Time	Source	Event ID	Task Ca...
Audit...	4/9/2026 7:39:21 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:36:31 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:31:33 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:25:58 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:24:21 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:24:05 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:17:27 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:17:07 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:17:06 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:16:31 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:12:25 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:35 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:28 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:26 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:24 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:18 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:18 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:03 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:01 PM	Micros...	4624	Logon
Audit...	4/9/2026 7:11:01 PM	Micros...	4624	Logon

Event 4624, Microsoft Windows security auditing.

General Details

An account was successfully logged on.

Subject:

Security ID: SYSTEM

Log Name: Security

Source: Microsoft Windows security ; Logged: 4/9/2026 7:39:21 PM

Event ID: 4624 Task Category: Logon

Level: Information Keywords: Audit Success

Results of filtered event ID: 4624

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 26,570

Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 2

Keyword...	Date and Time	Source	Event ID	Task Ca...
Audit...	4/9/2026 7:48:10 PM	Micros...	4625	Logon
Audit...	4/9/2026 7:48:06 PM	Micros...	4625	Logon

Event 4625, Microsoft Windows security auditing.

General Details

An account failed to log on.

Subject:

Security ID: SYSTEM

Log Name: Security

Source: Microsoft Windows security ; Logged: 4/9/2026 7:48:10 PM

Event ID: 4625 Task Category: Logon

Level: Information Keywords: Audit Failure

Results of filtered event ID: 4625

Security Number of events: 26,622

Filtered: Log: Security; Source: ; Event ID: 4648. Number of events: 118

Keyword...	Date and Time	Source	Event ID	Task Ca...
Audit...	25/8/2026 12:22:52 PM	Micros...	4648	Logon
Audit...	25/8/2026 11:00:54 AM	Micros...	4648	Logon
Audit...	25/8/2026 11:00:38 AM	Micros...	4648	Logon
Audit...	25/8/2026 12:22:52 PM	Micros...	4648	Logon
Audit...	26/8/2026 3:38:24 AM	Micros...	4648	Logon
Audit...	25/8/2026 7:53:18 PM	Micros...	4648	Logon
Audit...	25/8/2026 7:50:42 PM	Micros...	4648	Logon
Audit...	24/8/2026 12:13:39 AM	Micros...	4648	Logon

Event 4648, Microsoft Windows security auditing.

GeneralDetails

Subject:

Security ID:SYSTEM
Account Name:AZRA\$
Account Domain:WORKGROUP
Logon ID:0x3E7
Logon GUID:{00000000-0000-0000-0000-000000000000}

Account Whose Credentials Were Used:

Account Name:UMFD-9
Account Domain:Font Driver Host
Logon GUID:{00000000-0000-0000-0000-000000000000}

Target Server:

Target Server Name:localhost
Additional Information:localhost

Log Name:Security

Source:Microsoft Windows security ; Logged:25/8/2026 12:22:52 PM

Event ID:4648Task Category:Logon

Level:InformationKeywords:Audit Success

User:N/AComputer:Azra

OpCode:Info

Results of filtered event ID: 4648

INVESTIGATION STEPS

Step 1 - Filter for Failed Logons (Event ID 4625)

In Event Viewer, filter for Event ID 4625. For each event, record: the target username, the failure reason, the source network address, and the timestamp. Multiple 4625 events for the same username within a short period may indicate a brute-force attempt.

Security

Number of events: 26,578

Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 2

Keywords	Date and Time	Source	Event ID	Task Category
Audit	4/9/2026 7:48:06 PM	Micros	4625	Logon
Audit	4/9/2026 7:48:10 PM	Micros	4625	Logon

Event 4625, Microsoft Windows security auditing.

General

Details

An account failed to log on.

Subject:

Security ID: SYSTEM

Account Name: AZRA\$

Account Domain: WORKGROUP

Logon ID: 0x3E7

Logon Type: 2

Account For Which Logon Failed:

Security ID: NULL SID

Account Name: Azra

Account Domain: AZRA

Failure Information:

Failure Reason: Unknown user name or bad password.

Status: 0xC000006D

Sub Status: 0xC000006A

Process Information:

Caller Process ID: 0x0

Log Name: Security

Source: Microsoft Windows security ;

Logged: 4/9/2026 7:48:06 PM

Event ID: 4625

Task Category: Logon

Level: Information

Keywords: Audit Failure

User: N/A

Computer: Azra

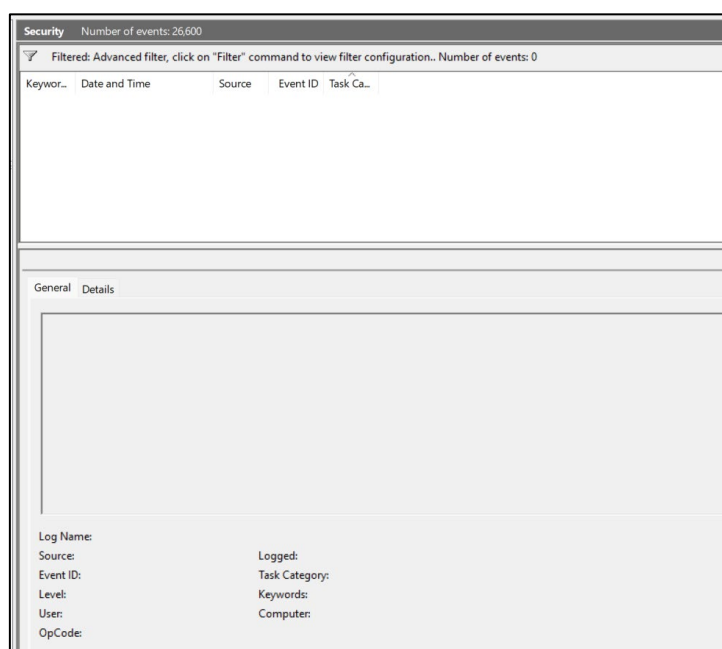
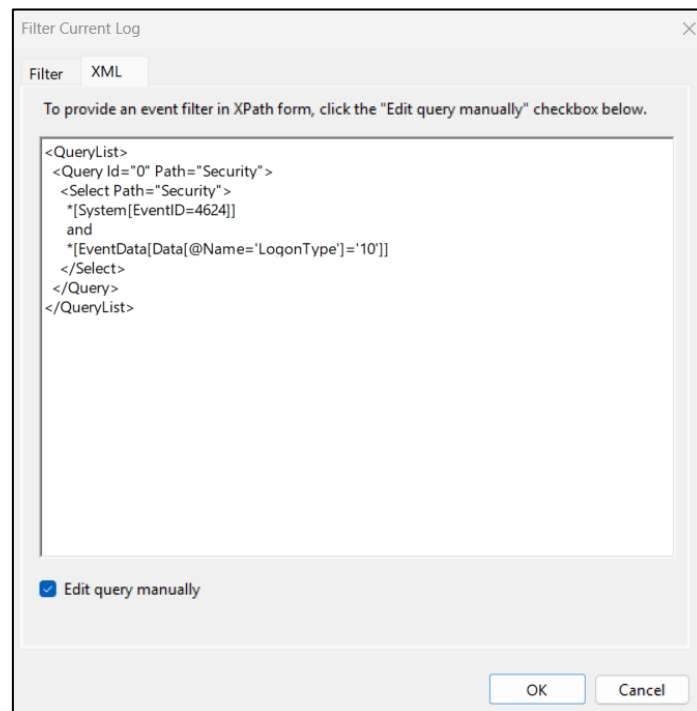
OpCode: Info

Step 2 - Check for Account Lockouts (Event ID 4740)

Filter for Event ID 4740. If present, note which account was locked out and the Caller Computer Name (the machine that triggered the lockout). Cross-reference with 4625 events to identify the source of repeated failures.

Step 3 - Review Successful Logons (Event ID 4624)

Filter for Event ID 4624. Focus on Logon Type 10 (RemoteInteractive/RDP) and Logon Type 3 (Network). These are more notable than Type 2 (interactive local logon). Note any unusual source IPs or off-hours logon times.



Logon type 10 and type 3 shows 0 results.

Step 4 - Check for Logon with Explicit Credentials (Event ID 4648)

Filter for Event ID 4648. Review two fields: the Subject Account (who initiated the logon) and the Account Whose Credentials Were Used (whose credentials were passed). A normal machine account will show a computer name ending in \$, this is expected Windows behaviour and not a concern. The risk indicator is a real user account authenticating to a different machine, which can signal lateral movement. If found, cross-reference with 4624 Type 3 events on the target host to confirm.

The screenshot shows the Windows Security Event Viewer interface. At the top, it says 'Security' and 'Number of events: 26,622'. Below that, a filter is applied: 'Filtered: Log: Security; Source: ; Event ID: 4648. Number of events: 118'. A list of events is shown, all with Event ID 4648 and Task Category 'Logon'. The details pane for Event 4648 is expanded, showing the following information:

- Subject:**
 - Security ID: SYSTEM
 - Account Name: AZRA\$
 - Account Domain: WORKGROUP
 - Logon ID: 0x3E7
 - Logon GUID: {00000000-0000-0000-0000-000000000000}
- Account Whose Credentials Were Used:**
 - Account Name: UMFD-9
 - Account Domain: Font Driver Host
 - Logon GUID: {00000000-0000-0000-0000-000000000000}
- Target Server:**
 - Target Server Name: localhost
 - Additional Information: localhost

At the bottom, the event details are summarized:

- Log Name: Security
- Source: Microsoft Windows security ; Logged: 25/8/2026 12:22:52 PM
- Event ID: 4648 Task Category: Logon
- Level: Information Keywords: Audit Success
- User: N/A Computer: Azra
- OpCode: Info

Step 5 - Build an Investigation Timeline

Record all relevant events below in chronological order to tell the story of what happened during the investigation period.

Timestamp	Event ID	Username	Source Machine / IP	Logon Type	Outcome / Notes
04/09/2026 08:12:33	4624	Azra	AZRA	2 (Interactive)	SUCCESS - Morning interactive logon.
04/09/2026 13:45:19	4624	Azra	AZRA	7 (Unlock)	SUCCESS - Workstation unlock, normal midday activity.
04/09/2026 19:48:06	4625	Azra	AZRA	2 (Interactive)	FAIL - Wrong password (0xC000006A). Attempt 1.
04/09/2026 19:48:10	4625	Azra	AZRA	2 (Interactive)	FAIL - Wrong password (0xC000006A). Attempt 2 (4 sec later).
04/09/2026 19:48:24	4624	Azra	AZRA	2 (Interactive)	SUCCESS - Correct password entered after 2 failed attempts.

FINDINGS

Failed Logon Activity

Two failed logon attempts (Event ID 4625) were identified for account 'Azra' on 4 September 2026 at 19:48:06 and 19:48:10, both from the local machine (AZRA) with Logon Type 2 (interactive). The Sub Status field in both events recorded 0xC000006A (Wrong Password), confirming the username was correct but the password was entered incorrectly. The two attempts were 4 seconds apart; a successful logon followed 14 seconds later at 19:48:24. This timing pattern is consistent with a user mistyping their password rather than an automated brute-force attempt. No other 4625 events were identified during the investigation period.

Account Lockout Activity

No account lockout events (4740) were identified during the investigation period.

Notable Successful Logons

All successful logon events (4624) during the investigation period were Logon Type 2 (Interactive) or Logon Type 7 (Workstation Unlock), originating exclusively from the local machine (AZRA). No Logon Type 10 (RemoteInteractive/RDP) or anomalous Type 3 (Network) logons from unexpected source IPs were detected. All timestamps fall within expected active hours. Logon activity is consistent with normal personal device usage.

ASSESSMENT

Classification

Failed logons observed [4625]: NORMAL – 2 failed attempts, correct password entered on next attempt. No brute-force indicators.

Account lockouts [4740]: N/A – No account lockout events were detected during the investigation.

Unusual successful logons [4624]: NORMAL – All logons are type 2 or type 7, local machine only within expected hours.

Explicit credential use [4648]: NORMAL – All 118 events attributed to Windows internal processes (Font Driver Host). No user-initiated explicit credential use detected.

Conclusion

The investigation of AZRA's Security logs did not identify any activity requiring escalation. The only anomalous events were two consecutive failed logon attempts (4625) on 4 September 2026, immediately followed by a successful authentication, consistent with a user mistyping their password, not an unauthorised access attempt.

KEY LEARNINGS

- Windows Security logs require filtering by Event ID to be useful because raw logs are too large to review manually.
- Event ID 4625 alone is not an alert, repeated 4625 events for the same user within seconds is the signal. However, if the failure count exceeds the threshold and no successful logon follows, it must be treated as a caution and reviewed further.
- Logon Type is critical: Type 10 (RDP) and Type 3 (network) carry more risk than Type 2 (local interactive) and type 7 (workstation unlock).
- 4648 events from SYSTEM using UMFD (Font Driver Host) accounts are normal Windows internals, not a risk. The signal is 4648 from a real user account authenticating to a different machine.